Matrix gradients and critical observations

Consolidated draft on signed channels and rectangular means

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Abstract

We consolidate finite matrix gradient estimates, spectral repair bounds and signed channel examples into one supporting appendix. An operator-valued Cwikel estimate gives an inverse moment bound with explicit counting-measure normalization. Finite-rank repair produces a capped moment criterion that admits zero and negative channels. A separate effective-resistance argument constructs stable signed families with uniform observation rank and penalty, and a small Schur matrix detects coupled instabilities. A critical rectangular mean obstruction is also transferred to actual flat coexact cochains. An explicit torus example shows that individually repairable channels need not remain repairable together by a scalar differentiable observation. The underlying spectral and resistance methods are classical. The statements concern the defined finite gradient form; they establish neither an interacting Yang–Mills form identity nor a continuum mass gap.

Status and provenance. This is a grouped appendix candidate for the existing integrated manuscript, not a separate deposit. Its analytical sources and finite evidence are pinned at repository commit `66580c59`; full identifiers and execution records are in Section 8. The published working paper v6 and its assessment report 154 remain separate artifacts. This new PDF has no inherited assessment score, independent review or Lean certificate. The parent is AIRR working paper ARR-2026-7EPM3V3DP796BBRR version 6; its unchanged PDF SHA256 is

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1 Actual finite objects and observation scope

Let $D \geq 3$, $n \geq 2$, $h \geq 1$, $V = n^D$, $r = Dh$, and $p = D/2$. On $X = (\mathbb{Z}/n\mathbb{Z})^D$ use the counting inner product. For $u : X \to \mathbb{R}^h$, define

$$du(x) = (u(x + e_i) - u(x))_{i=1}^{D}, \qquad Q_A(u) = \sum_x du(x)^T A(x) du(x), \quad L_A = d^* A d,$$

where $A(x)$ is a real symmetric $r \times r$ matrix. A channel is one local coefficient eigenvector, counted with multiplicity. Distinct matrices need not commute. All derivatives have unit spacing. A physical derivative nd requires an explicit rescaling of every energy and threshold.

The number $N_{<c}(L_A)$ counts all eigenvalues strictly below $c > 0$, including negative and zero ones, with ordinary real multiplicity. Spectral coordinates on this low space give a real observation C and penalty

$$\Phi(y) = \sum_{\lambda<c} (c - \lambda) y_\lambda^2, \qquad Q_A(u) + \Phi(Cu) \geq c\|u\|^2. \tag{1}$$

This is an existence result for an actual, generally nonlocal observation. It does not construct the unresolved local rectangular observation. For signed forms, $c\|y\|^2$ alone need not be sufficient.

2 The positive matrix estimate

Put

$$\alpha_D = \frac{D}{2}\left(\frac{D}{D-2}\right)^{D-1}\left(\frac{3}{4}\right)^{D}.$$

Theorem 1. *If every $A(x)$ is positive definite, then*

$$N_{<c}(L_A) \le h + \alpha_D c^p \sum_x \operatorname{tr} A(x)^{-p}. \tag{2}$$

Proof. On the mean-zero space H_0, write $L_0 = d^*d$ and $J = dL_0^{-1}$. For any $b \in H_0$, the flow $j_* = AdL_A^{-1}b$ has $d^*j_* = b$. Every other feasible flow $j = j_* + k$ has $d^*k = 0$, and expansion gives

$$\langle j, A^{-1}j\rangle = \langle b, L_A^{-1}b\rangle + \langle k, A^{-1}k\rangle.$$

Taking the actual feasible flow Jb proves $L_A^{-1} \le J^*A^{-1}J$. No coefficients commute in this argument.

The unitary counting Fourier transform on r components has $L_1(\ell_2^r) \to L_\infty(\ell_2^r)$ norm $V^{-1/2}$. The symbol of J is $z(k)/\lambda(k)$ with $z_i(k) = e^{2\pi i k_i/n} - 1$ and $\lambda(k) = 4\sum_i \sin^2(\pi k_i/n)$; set $J(0) = 0$. It is rectangular $r \times h$. Embed $\mathbb{R}^h$ isometrically in $\mathbb{R}^r$ by I and use the square symbol $G(k) = J(k)I^*$. Thus $\|G(k)\| = \lambda(k)^{-1/2}$ for $k \neq 0$. Writing $\mathcal{F}_r$ for the counting transform, the actual factorization is $J = \mathcal{F}_r^* M_G \mathcal{F}_r I$. Centered frequency representatives b satisfy $\lambda(k) \ge 16|b|^2/n^2$. Counting the cube containing $0 < |b| < n/(4t)$ gives

$$\sup_{t>0} t^D \#\{k : \|G(k)\| > t\} \le (3/4)^D V.$$

When the radius is at most one the set is empty; otherwise $(2\lfloor R\rfloor + 1)^D \le (3R)^D$ suffices.

Apply Frank's operator-valued Cwikel theorem [1, Theorem 3.1] with $q = D$, $f = A^{-1/2}$, this square multiplier and counting measures. Its constant is $\frac{D}{2}(\frac{D}{D-2})^{D-1}V^{-1}$, so the displayed frequency V cancels exactly once. It bounds the number of singular values of $A^{-1/2}J$ strictly above s by $\alpha_D s^{-D}\sum_x \operatorname{tr} A(x)^{-p}$. The right factor $\mathcal{F}_r I$ is an isometry and cannot increase singular-value counts. Apply min-max to the inverse comparison at threshold $1/c$ on H_0, then add the h constant modes. Complexification of a real symmetric matrix preserves multiplicities. □

Input and interpretation. The operator-valued estimate is classical; this proof fixes its finite normalization, padding and actual form dictionary. It is not a new general Cwikel theorem. A matrix arithmetic mean alone does not imply its inverse moment hypothesis. For the formal flat cochain application one may set $h = D(N_c^2 - 1)$, but an interacting Hessian or weighted curl/divergence form must first be proved equal or comparable to this Q_A.

3 Spectral repair and the capped budget

Theorem 2. *Let $A(x)$ now be arbitrary real symmetric matrices. In local orthonormal eigenbases choose a set S of channels containing all nonpositive ones. Let P_S be their orthogonal projection and $R(S) = \operatorname{rank}(P_S d)$. Then*

$$N_{<c}(L_A) \le \min\left\{hV,\ h + R(S) + \alpha_D c^p \sum_{j\notin S} \lambda_j^{-p}\right\}. \tag{3}$$

Writing ν_0 for the total multiplicity of nonpositive local eigenvalues, one consequently has

$$N_{<c}(L_A) \le \min\{hV,\ h + F_c(A)\}, \quad F_c(A) = \nu_0 + \sum_{\lambda_j>0} \min\{1, \alpha_D c^p \lambda_j^{-p}\}. \tag{4}$$

Proof. Replace selected eigenvalues by T larger than zero and all of them. The resulting field A_T is positive definite and

$$L_{A_T} - L_A = d^* P_S (T\mathrm{I} - A) P_S d \geq 0, \quad \operatorname{rank}(L_{A_T} - L_A) = R(S).$$

The rank equality follows because $T\mathrm{I} - A$ is strictly positive on $\operatorname{im} P_S$. Finite-rank min-max gives $N_{<c}(L_A) \leq N_{<c}(L_{A_T}) + R(S)$: intersect the low space of L_A with the kernel of the update and the orthogonal complement of the low space of L_{A_T}. A dimension larger than $R(S)$ would give a nonzero vector whose identical quadratic forms are both strictly below and at least $c\|u\|^2$. Theorem 1 and $T \to \infty$ prove (3). For (4), use $R(S) \leq |S|$ and select all positive channels with $\alpha_D c^p \lambda_j^{-p} \geq 1$ as well. □

A strict extension of the full moment criterion. Let $m \geq 1$ and $T_m = \max\{1, V/(1+m)^p\}$. If $F_c(A) \leq KT_m$, then $N_{<c}(L_A) \leq (h+K)T_m$. Every former SPD budget

$$\frac{1}{rV} \sum_x \operatorname{tr}((mA(x)^{-1})^p) \leq M$$

implies this capped budget, with $K = 2^p \alpha_D c^p r M$. Conversely choose $m = n^2$, one actual diagonal edge channel $\epsilon = n^{-4}$, and all others m. Now $T_m = 1$ and $F_c(A) \leq 1 + \alpha_D c^p r$, whereas the normalized full inverse moment is at least n^{2D}/r and diverges. These bounds are derived for this explicit family; no new moment bound is assumed.

Sparse contamination and its limitation. Selecting every channel with $\lambda < m$ gives

$$N_{<c}(L_A) \leq h + R_m + \alpha_D c^p r V m^{-p}, \qquad R_m = \operatorname{rank}(\mathbf{1}_{(-\infty,m)}(A)d).$$

An actual selected-channel count $k_m \leq KT_m$ suffices since $R_m \leq k_m$. If $A(x) \geq m\mathrm{I}$ outside a specified set B, then $k_m \leq r|B|$. Neither condition follows from an arithmetic mean. This particular prescribed-threshold count criterion does not subsume every full moment family: $A = (m/2)\mathrm{I}$ has normalized inverse moment 2^p but every channel lies below m. The two particular sufficient criteria are incomparable; the general capped criterion (4) does include the old one.

Actual rank and unavoidable costs. For diagonal direction/component channels the selected rows are graph incidences. If κ_a counts all components, including isolated vertices, of the selected-edge graph for component a, then $R(S) = \sum_a (V - \kappa_a)$: its kernel consists precisely of fields constant on those components. Parallel rows at $n = 2$ do not change that kernel. This formula is not applied to rotated local eigenchannels. Isolating q nonadjacent vertices by zeroing their incident edges gives at least $q+1$ kernel modes. Replacing these zeros by $0 < \epsilon < c/(2D)$ yields a q-dimensional strict low subspace of delta functions. The defect-count cost cannot simply be deleted.

In (1), a generic bound is $K \leq c + 4Db$ with $b = \max_x \|A_-(x)\|$, since $\|du\|^2 \leq 4D\|u\|^2$. One negative channel $-B$ with all other channels m, at $n \geq 3, h = 1$, has the trial vector $v = \delta_y - \delta_x$ on that edge and

$$\frac{Q_A(v)}{\|v\|^2} = -2B + (2D-1)m.$$

Any normalized linear observation with penalty bounded by $K\|y\|^2$ therefore needs $K \geq c + 2B - (2D-1)m$. Sparsity alone does not bound penalty.

4 Stable signed families from actual resistance

This section is independent of the Cwikel input. Its mechanism is classical effective resistance and finite-rank congruence [2, 3]. Let e be one actual direction/component row and $v = d^*e$. With all coefficients m except this one set to $-B$, $B \geq 0$,

$$H = mL_0 - (m + B)vv^*, \qquad L_0 = d^*d.$$

On H_0 let $G = L_0^{-1}$. Translations and coordinate permutations make every scalar row leverage equal. Their sum is $\operatorname{tr}(dGd^*) = V - 1$, so

$$\rho = v^*Gv = \frac{V - 1}{DV}. \tag{5}$$

Components decouple; the same value holds for every h. The DV rows are counted individually even at $n = 2$.

Proposition 3. *The exact threshold is $B_* = m(DV/(V - 1) - 1)$. On H_0, H is positive definite for $B < B_*$, has one zero direction at equality, and has exactly one negative direction above it. If $m = n^2$, $B = \theta m$ and $0 \leq \theta < D - 1$, its mean-zero gap is at least*

$$g = 16(D - 1 - \theta)/D > 0.$$

For every $0 < c \leq g$, the normalized mean observation $C(u) = V^{-1/2} \sum_x u(x) \in \mathbb{R}^h$ and $\Phi(y) = c\|y\|^2$ give coercivity with optimal rank h, uniformly in n.

Proof. Congruence by $G^{1/2}$ gives $m\mathrm{I} - (m + B)ww^*$ with $\|w\|^2 = \rho$, proving the phase claim. The equality witness is Gv. Cauchy–Schwarz in the L_0 energy gives $|v^*u|^2 \leq \rho\langle u, L_0u\rangle$, hence $H \geq [m - (m + B)\rho]L_0$ on H_0; equality holds on Gv. For the stated scaling the coefficient is at least $m(D - 1 - \theta)/D$. The actual first nonconstant Fourier eigenvalue of L_0 is $4\sin^2(\pi/n) \geq 16/n^2$. This proves the gap. Decompose into constants and H_0 to prove the displayed observation inequality. The constant kernel forces rank at least h. □

For $\theta > 0$, negative coefficient depth grows as θn^2, yet the penalty is uniformly c. A bounded-depth hypothesis is sufficient for the generic estimate, not necessary for every stable signed family. The actual form is positive despite an indefinite coefficient field; this does not contradict the fixed-m, $B \to \infty$ penalty obstruction above.

5 Coupled channels and a nonlinear observation obstruction

Choose q actual rows of d as Z, and $W = \operatorname{diag}(w_j) > 0$. The modified form is exactly $H = mL_0 - Z^*WZ$. Define

$$\Gamma = ZGZ^*, \qquad \mathcal{C} = mW^{-1} - \Gamma, \quad N_0 = h(V - 1).$$

Schur elimination of either positive diagonal block in $\left[\begin{smallmatrix} mL_0 & Z^* \\ Z & W^{-1} \end{smallmatrix}\right]$ on $H_0 \oplus \mathbb{R}^q$ proves

$$n_-(H) = n_-(\mathcal{C}), \quad n_0(H) = h + n_0(\mathcal{C}), \quad n_+(H) = N_0 - q + n_+(\mathcal{C}). \tag{6}$$

Indeed the two congruences give the identity

$$\operatorname{inertia}(H|_{H_0}) + (q, 0, 0) = (N_0, 0, 0) + \operatorname{inertia}(\mathcal{C}/m).$$

No disjoint-cycle or independent-row assumption occurs. In particular, H is positive semidefinite with exactly h constant zeros if and only if $\mathcal{C}$ is positive definite. General signed matrix criteria preexist in [4]; our Γ uses the positive reference inverse.

Example 4. *Take $D = 3, n = 2, h = 1, m = 4$, and change the first two direction rows at one vertex separately or together to coefficient $-84/11$. Then*

$$\Gamma = \begin{pmatrix} 7/24 & 5/48 \\ 5/48 & 7/24 \end{pmatrix}, \quad w = 128/11, \quad w/m = 32/11.$$

The diagonal follows from (5)*; the off-diagonal Walsh sum is* $(1/8)(1/2 + 1/3) = 5/48$*, with both forward orientations counted. Each single channel is below its critical magnitude* $68/7$ *and has full inertia* $(7, 0, 1)$*. Together the reduced matrix has diagonal* $5/96$ *and off-diagonal* $-5/48$*, so its eigenvalues are* $-5/96, 5/32$*. Equation* (6) *gives joint inertia* $(6, 1, 1)$*.*

Each single form has mean-zero gap at least $16(1 - (32/11)(7/24)) = 80/33 > 1$*. The same scalar mean observation with* $\Phi(y) = |y|^2$ *repairs either single form at* $c = 1$*. The joint form cannot be repaired by any scalar observation* $F : \mathbb{R}^8 \to \mathbb{R}$ *differentiable at zero with* $F(0) = 0$ *and* $0 \le \Phi(z) \le K|z|^2$ *near zero. Its negative eigenvector and constant vector span a two-dimensional nonpositive space. The derivative of* F *has a nonzero kernel vector* v *there. Then* $F(tv) = o(t)$ *and* $\Phi(F(tv)) = o(t^2)$*, whereas* $Q_A(tv) \le 0$*, contradicting coercivity at* $c = 1$ *for small nonzero* t*.*

This refutes the shortcut from individual to joint control, not the full matrix resistance criterion or any theorem retaining a disjoint-cycle hypothesis. The q-row perturbation also gives $N_{<c}(H) \le h + q$ for $0 < c \le 16m/n^2$. Extra zero or negative modes provide the necessary rank lower bound in (6); they do not alone bound the penalty coefficient.

6 Critical constants for rectangular mean observations

For a disjoint partition $\mathcal{P}$ of a finite counting space, its normalized means are $(Cf)_R = |R|^{-1/2} \sum_{x \in R} f(x)$. They satisfy $CC^* = \mathrm{I}$ and $C^*C = P$, the orthogonal projection onto functions constant on each cell. This concrete normalization matters for the penalty.

Theorem 5. *Let $L = L^*$ on a finite real Hilbert space, $c > 0$, and C a coisometry onto $\mathbb{R}^q$, $q \ge 1$. On $\mathcal{K} = \ker C$ and $\mathcal{O} = \operatorname{im} C^*$, write*

$$L - c\mathrm{I} = \begin{pmatrix} A_0 & X \\ X^* & B_0 \end{pmatrix}.$$

A finite $k \ge 0$ satisfying $L + kC^*C \ge c\mathrm{I}$ exists if and only if

$$A_0 \ge 0, \qquad X^*(\ker A_0) = 0. \tag{7}$$

When these hold, its smallest value is

$$k_{\min} = \max\{0, \lambda_{\max}(X^* A_0^\dagger X - B_0)\}.$$

Here $A_0^\dagger$ is the Moore–Penrose inverse on $\mathcal{K}$.

Proof. The restriction to $\mathcal{K}$ forces $A_0 \ge 0$. For $z \in \ker A_0$ the augmented form at $tz + y$ has linear term $2t\langle z, Xy\rangle$ and no quadratic term in t. Nonnegativity for every real t forces $X^*z = 0$. This is equivalent to $\operatorname{im} X \subseteq \operatorname{im} A_0$. Under that condition the full form at $z + y$ is exactly

$$\langle z + A_0^\dagger Xy, A_0(z + A_0^\dagger Xy)\rangle + \langle y, (B_0 + k\mathrm{I} - X^* A_0^\dagger X)y\rangle.$$

Finite spectral decomposition proves the criterion and the minimum. □

This is the classical generalized Schur criterion [5, Theorem 1], applied to actual means; it is not a new general matrix theorem. Strict positivity of A_0 suffices, whereas a weak bound at the target requires the extra compatibility in (7).

Example 6. *On* $(\mathbb{Z}/2\mathbb{Z})^3$ *take the same scalar coefficient in every gradient direction,*

$$a(x) = 6 + (-1)^{x_1} + 4(-1)^{x_2}, \qquad L = d^* a d.$$

Its values are $1, 3, 9, 11$*, each twice, and its mean is* $m = 6$*; hence* $1 \le a \le 2m$*. Choose the four actual rectangles* $R_{yz} = \{0,1\} \times \{y\} \times \{z\}$ *and* $(Cf)_{yz} = (f(0,y,z) + f(1,y,z))/\sqrt{2}$*. The best coercivity constant of* L *on* $\ker C$ *is* 16*, but no finite quadratic penalty for these means attains the same constant on the full space. Every* $0 < c < 16$ *is attainable. Thus the supremum* 16 *is not attained.*

Proof. For $S \subseteq \{1,2,3\}$ let $\chi_S(x) = (-1)^{\sum_{i \in S} x_i}/\sqrt{8}$. These are an orthonormal real Walsh basis. Retain all 24 oriented forward rows, including opposite rows at period two. The actual adjoint calculation gives

$$\begin{aligned} L\chi_S = 24|S|\chi_S &+ 4\big(|S| - \mathbf{1}_{1 \in S}\big)\chi_{S \triangle \{1\}} \\ &+ 16\big(|S| - \mathbf{1}_{2 \in S}\big)\chi_{S \triangle \{2\}}. \end{aligned}$$

Indeed $d_i \chi_S = -2\chi_S$ when $i \in S$, and the corresponding adjoint term is $2(a + T_i a)\chi_S$; absent directions contribute zero. In the kernel basis $(\chi_1, \chi_{12}, \chi_{13}, \chi_{123})$ the compression has the two blocks

$$\begin{pmatrix} 24 & 16 \\ 16 & 48 \end{pmatrix}, \qquad \begin{pmatrix} 48 & 32 \\ 32 & 72 \end{pmatrix}.$$

After subtracting 16I, the first is positive semidefinite with kernel vector $(-2, 1)$, and the second is positive definite, with first pivot 32 and determinant 768. This proves the complete kernel bound, not just a bound on selected test vectors.

Set $v = -2\chi_1 + \chi_{12}$ and $u = \chi_2$. Direct evaluation gives

$$Cv = 0, \quad \|v\|^2 = 5, \quad \|Cu\|^2 = \|u\|^2 = 1, \quad (L - 16\mathrm{I})v = 4u, \quad \langle u, (L - 16\mathrm{I})u \rangle = 8.$$

The augmented form on $u + tv$ is therefore $8 + 8t + k$, negative for $t < -(8+k)/8$. No finite k works. For $c < 16$ the compressed block is strictly positive and Theorem 5 supplies a finite penalty; for $c > 16$, v violates coercivity already in the kernel. □

A quantitative penalty obstruction. Write $\delta = 16 - c > 0$. On the orthonormal pair $(v/\sqrt{5}, u)$ the augmented form has matrix

$$\begin{pmatrix} \delta & 4/\sqrt{5} \\ 4/\sqrt{5} & 8 + \delta + k \end{pmatrix}.$$

Its determinant gives the necessary bound

$$k \ge \max\{0, 16/(5\delta) - 8 - \delta\}. \tag{8}$$

This lower bound need not equal the full optimum. It proves divergence as the target approaches 16 from below without numerical extrapolation. If $F(0) = 0$, $DF(0) = C$ and $0 \le \Phi(y) \le k\|y\|^2$ near zero, local coercivity would imply $L + kC^*C \ge c\mathrm{I}$ by substituting sf, dividing by s^2 and taking $s \to 0$. Higher order terms with this same derivative do not repair the failure at $c = 16$.

The exact penalty rather than only a lower bound. For this same scalar example, the full optimum at $c = 16 - \delta$, $0 < \delta < 16$, is

$$k_{\min}(c) = \max\left\{c, \frac{16(8+\delta)}{\delta(40+\delta)} - 8 - \delta\right\}. \tag{9}$$

Here is a complete eight-dimensional check. Order the kernel basis as $(\chi_1, \chi_{12}) \oplus (\chi_{13}, \chi_{123})$ and the observed basis as $(\chi_\varnothing, \chi_2) \oplus (\chi_3, \chi_{23})$. At target 16 the kernel, observed and cross blocks of $L - 16\mathrm{I}$ split into

$$A_1 = \begin{pmatrix} 8 & 16 \\ 16 & 32 \end{pmatrix}, \quad B_1 = \mathrm{diag}(-16, 8), \quad X_1 = \begin{pmatrix} 0 & 0 \\ 0 & 4 \end{pmatrix},$$
$$A_2 = \begin{pmatrix} 32 & 32 \\ 32 & 56 \end{pmatrix}, \quad B_2 = \begin{pmatrix} 8 & 16 \\ 16 & 32 \end{pmatrix}, \quad X_2 = \mathrm{diag}(4, 8).$$

At $c = 16 - \delta$, add $\delta\mathrm{I}$ to each A_i, B_i, leaving X_i unchanged. The first observed Schur block is diagonal, with entries

$$16 - \delta, \qquad \frac{16(8+\delta)}{\delta(40+\delta)} - 8 - \delta.$$

The second observed Schur block cannot exceed $(16 - \delta)\mathrm{I}$: $A_2 - 4\mathrm{I}$ has first pivot 28 and determinant 432, while $B_2 \geq 0$; hence $X_2^*(A_2 + \delta\mathrm{I})^{-1}X_2 \leq 16\mathrm{I}$. The exact Schur minimum in Theorem 5 therefore proves both necessity and sufficiency of (9), including all eight dimensions. The branch transition is

$$\delta_0 = \frac{\sqrt{3529} - 59}{3} : \quad k_{\min}(c) = c \text{ for } \delta \geq \delta_0,$$

and the other branch is larger for $0 < \delta < \delta_0$. Comparing the entries reduces to $3\delta^2 + 118\delta - 16 \geq 0$. Partial fractions give the sharper critical expansion

$$k_{\min}(c) = \frac{16}{5(16-c)} - \frac{192}{25} + O(16 - c) \qquad (c \uparrow 16). \tag{10}$$

The previous lower bound (8) alone is not the full optimum.

The general finite mechanism. For the blocks of Theorem 5 at the least kernel eigenvalue $c_* > 0$, let P project onto $\ker A_0$. At $c = c_* - \delta$,

$$k_{\min}(c) = \max\{0, \lambda_{\max}(X^*(A_0 + \delta\mathrm{I})^{-1}X - B_0 - \delta\mathrm{I})\}.$$

Finite spectral splitting writes the inverse as $\delta^{-1}P$ plus a bounded remainder converging to $A_0^\dagger$. The variational formula for the largest eigenvalue consequently gives

$$\lim_{\delta \downarrow 0} \delta k_{\min}(c_* - \delta) = \|PX\|^2.$$

If $PX \neq 0$, the penalty diverges as $\|PX\|^2/\delta + O(1)$ and the critical target fails. If $PX = 0$, it converges to the finite critical Schur minimum. This uses classical finite Schur and resolvent principles, not a claim of a new general matrix theorem.

7 Transfer to actual flat coexact cochains

The preceding obstruction also holds on the actual finite coexact domain, with a proved form dictionary. Fix $N_c \geq 2$ and an actual coordinate unit $e \in \mathbb{R}^{N_c^2-1}$. At the trivial background, a physical one-cochain is $A_j(x) \in \mathbb{R}^{N_c^2-1}$, $j = 1, 2, 3$, with counting inner product. Use one unordered plaquette copy for each $i < j$:

$$(d_1 A)_{ij} = d_i A_j - d_j A_i, \qquad \operatorname{div} A = \sum_j (A_j - T_j^{-1} A_j).$$

Expansion of curl squares and cyclic summation by parts cancel the cross terms against divergence, proving the all-cochain identity

$$\|d_1 A\|^2 + \|\operatorname{div} A\|^2 = \sum_{i,j,x} \|d_i A_j(x)\|^2.$$

These are the actual repository conventions, not a supplied comparison. Take the actual componentwise repair coefficient $w(i,(x,j)) = a(x) - 1 \geq 0$. The total Hodge-plus-repair form is consequently

$$\begin{aligned} Q_w(A) &= \|d_1 A\|^2 + \|\operatorname{div} A\|^2 + \sum_{i,j,x} w(x) \|d_i A_j(x)\|^2 \\ &= \sum_{i,j,x} a(x) \|d_i A_j(x)\|^2. \end{aligned} \tag{11}$$

It is a direct sum of $3(N_c^2 - 1)$ scalar copies. Componentwise normalized means on the same four bars have output dimension $12(N_c^2 - 1)$, and the whole ambient kernel has coercivity constant 16 by the two blocks in Example 6. Restriction to coexact fields preserves the inequality; coisometry on the restricted domain is not assumed.

Proposition 7. *At target* 16*, no finite quadratic penalty for these actual componentwise rectangle means supplies coercivity even on* $\operatorname{im} d_1^*$*. The same is true for local nonlinear observations with derivative equal to those restricted means and a locally quadratic penalty. On that domain every* $0 < c < 16$ *is attainable, and the necessary bound* (8) *still applies.*

Proof. Construct actual fields $U_3 = ue$, $V_3 = ve$ from Example 6, with all other components zero. They are isometric copies of u, v and independent of x_3, so their actual backward divergence vanishes. More explicitly, use unordered two-cochains with only

$$B_{23}^U = -\tfrac{1}{2}\chi_2 e, \qquad B_{13}^V = (\chi_1 - \tfrac{1}{2}\chi_{12})e$$

nonzero. At side two $d_i^* = T_i^{-1} - \mathrm{I} = T_i - \mathrm{I}$. The adjoint of the displayed curl puts $d_i^* B_{ij}$ in component j and $-d_j^* B_{ij}$ in component i. Since $d_3^* B^U = d_3^* B^V = 0$ and each relevant character has multiplier -2, these actual potentials satisfy $d_1^* B^U = U$ and $d_1^* B^V = V$. Thus the entire line $U + tV$ is coexact.

Equation (11) and the actual means give the same augmented value $8 + 8t + k$. Scaling a violating coexact field towards zero also excludes locally bounded quadratic penalties and the stated nonlinear observations. Strict positivity on the ambient kernel supplies a finite penalty for every $c < 16$, which can then be restricted. The coexact field V excludes $c > 16$. The same two-dimensional determinant proves the bound. □

Scope and source of the transfer. The pinned repository gives the actual form identities in these two sources:

`componentwiseFlatGradient_eq_hodge`
`YangMills/RG/ComponentwiseFlatHodge.lean`

`actualTotalStiffness_energy`
`YangMills/RG/ActualPacketSupportEnvelope.lean`

Their proofs and the actual curl/divergence definitions were read; no new Lean compilation is claimed. Unit 166 checks the newly constructed physical stencils rather than relabelling a scalar diagnostic. For derivatives nd_i at $n = 2$, energies and targets scale by four, so 16 becomes 64 with rescaled potentials. This is finite rescaling, not a physical mass-gap claim.

The example has finite inverse moments at this one size. It refutes weak kernel sufficiency for these prescribed rectangles at the same target; it does not rule out another rectangular partition with the desired rank, or establish an unbounded-size violation of the critical rank-order conjecture. Only the trivial-background form has been identified here.

The sharp coexact leading coefficient. Let $k_{\mathrm{coex}}(c)$ be the minimum penalty for the actual means restricted to $\operatorname{im} d_1^*$. The witness pair just proved and the ambient direct-sum identity give, for $\delta = 16 - c \in (0, 16)$,

$$\max\{0, 16/(5\delta) - 8 - \delta\} \le k_{\mathrm{coex}}(c) \le k_{\min}(c).$$

Multiplying by δ and using (9) proves

$$\lim_{c \uparrow 16} (16 - c) k_{\mathrm{coex}}(c) = \frac{16}{5}.$$

No coisometry is assumed for the restricted mean map. This proves the same leading coefficient on the actual coexact domain, not equality of the full ambient and restricted optima or of their next constant terms.

8 Reproduction and limitations

The source repository is THE ERIKSSON PROGRAMME PR186, stacked on PR185. All paths below are under `reproducibility/matrix-moment-sweep-20261106/`. Full source for this consolidation is pinned at

`66580c59ea9ef5f3c0374b339a165c9f89173db8`.

Units 162 and 163 each used two distinct own CPU allocations, four actual fresh sparse Git clones at one pinned source, and eight normal/optimized Python processes. Outputs matched within each unit. Both runtimes were deleted and disconnected after evidence export. Full stdout, stderr, exit sentinels, Git logs, exact source bytes and manifests are in the corresponding `evidence162-*.zip` and `evidence163-*.zip` archives.

Unit	Executed source prefix	Fixture cases	Mutations per process
162	`f644c093`	four	seven types
163	`378e879b`	four	seven types across four cases

The 163 fixtures are $(D, n, h) = (3, 2, 1), (3, 3, 1), (3, 3, 2), (4, 2, 2)$. They check actual Green and leverage identities, below/equal/above phase inertia, equality witnesses, gap and mean-observation coercivity, and coupled pair inertia. Unit 162 covers zero and near-zero stars, one large negative channel and rationally rotated local eigendirections. Every acceptance check raises explicitly on failure in both ordinary and `-O` execution. There are 28 fixture-mutation pairs per 163 process, not 28 distinct types. The two unit 163 recipe times were 53.83 and

29.98 seconds; requested-to-release times were 179.71 and 102.74 seconds. Billed compute units were not exposed.

Units 165 and 166 each used two distinct own CPU allocations, four actual fresh pinned Git clones and eight ordinary/optimized processes. Unit 165 checks the entire scalar torus matrix, the rectangular projection and three distinct mutations per process. Unit 166 constructs the actual 72×72 physical matrix at $D = 3, n = 2, N_c = 2$, with 72 unordered curl rows, 24 backward divergence rows and 216 gradient rows. It checks the full Hodge-plus-repair identity, both actual adjoint potentials, divergence zero and the rectangular witnesses. Four distinct physical mutations are rejected per process: omit divergence, double the unordered curl, reverse a potential sign and put the field in the wrong component. Both 166 archives have 55 verified manifest subjects. Their recipe times are 5.28 and 8.19 seconds; requested-to-release times are 46.23 and 49.10 seconds. Both runtimes were disconnected after export. Billed compute units remain unknown. The analytic all-N_c coordinate embedding and nonlinear limit are not extra executed cases.

Finite replay checks these finite matrices, not the universal analytical statements. Arbitrary q, unequal weights, the all-size capped-family example and the nonlinear derivative consequence are analytical arguments, not extra executed fixtures. Earlier 159 diagnostics used source downloads rather than fresh Git clones; their original limitation remains in their own records. No Lean compilation certifies this appendix. A full independently prepared audit of the imported Fourier adapter and the application form dictionary remains necessary before broader claims.

Unit 168 derives the exact penalty and critical residue analytically. Unit 169 then builds the actual scalar spatial matrix from all 24 forward rows and the actual pair-average projection using rational arithmetic. Generic full 8×8 positive-semidefinite elimination checks feasibility; separate rational row reduction finds observed null vectors, which prove that every strictly smaller penalty fails. Thus excess feasible penalties are rejected rather than labelled optimal. Nine distinct rational δ values cover both branches and bracket the transition without rounding. Six mutation types per process reject the old lower bound used as the optimum, an omitted constant-mode floor, doubled residue, excess penalty, erased actual coupling and collapsed opposite rows. Two own CPU allocations, four fresh pinned Git clones and eight ordinary or optimized processes give identical payloads. Each complete archive has 53 verified manifest subjects and 16 real zero-exit sentinels. Recipe times were 3.99 and 3.36 seconds; requested-to-release times were 36.32 and 41.92 seconds, including provisioning and export. Both runtimes were disconnected after export; billed units are unknown. Complete records are in `evidence169-*.zip` and `reproduction169-summary.json`. These finite checks do not certify every real δ, a new physical 72×72 replay, the universal resolvent lemma or the coexact limiting argument. Those remain the displayed analytic proofs. No new Lean build certifies these extensions.

9 Model assessments and review context

This appendix was prepared and checked analytically in the continuing Codex session that prepared its sources and diagnostics. The application does not expose the exact backend model identity for that preparation. There is no claim of independence, blinding, human peer review or editorial acceptance. This exact new PDF has no external model assessment and no numerical score. An assessment of an earlier PDF does not transfer to this artifact.

The retained objections are substantive: the Cwikel mechanism and Schur principles are classical, so independent originality is unestablished; the observation from spectral coordinates is generally nonlocal; sparse-count and full-moment criteria must not be conflated; a diagonal graph-rank formula does not cover rotated eigenchannels; negative depth is neither universally

dispensable nor universally necessary; the full off-diagonal interaction must be retained; critical kernel compatibility cannot be omitted at the target; finite diagnostics do not prove all-size statements; and no interacting physical form equality has been supplied. These limits are part of the result, not editorial acceptance conditions represented as satisfied.

The original positive critical rectangular construction remains open. The common-refinement route remains stopped after three unsuccessful units. The recorded PR29 baseline stays 20/41 with TermSource 0. This appendix adds supporting finite mathematical results and useful counterexamples without changing those records. It is a draft for consolidation, not a deposit, acceptance or new verified public release.

References

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[3] D. A. Spielman, *Spectral Graph Theory Lecture 14*, 15 October 2018, Claim 14.6.2. `https://www.cs.yale.edu/homes/spielman/561/lect14-18.pdf`.

[4] W. Chen, J. Liu, Y. Chen, S. Z. Khong, D. Wang, T. Başar, L. Qiu and K. H. Johansson, *Characterizing the positive semidefiniteness of signed Laplacians via effective resistances*, IEEE CDC (2016), pp. 985–990. `https://people.kth.se/~kallej/papers/network_cdc16chen.pdf`.

[5] N. J. Higham, *What Is the Schur Complement of a Matrix?*, 1 June 2023, Theorem 1, generalized Schur complement. `https://nhigham.com/2023/06/01/what-is-the-schur-complement-of-a-matrix/`.